

Ultraviolet Trajectories and Conditional Completion

Companion note: isolating UV identification inputs for the macroscopic Yang–Mills mass gap

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Abstract

Route 1 establishes a finite-depth entrance mechanism for four-dimensional lattice Yang–Mills at large inverse coupling β : after a computable blocking depth, the exact blocked specification enters a uniform Dobrushin contraction basin, yielding infinite-volume uniqueness and exponential Euclidean clustering for bounded gauge-invariant observables at a fixed macroscopic scale $\ell > 0$. To express these Euclidean estimates as a continuum spectral statement one must additionally specify an ultraviolet (UV) trajectory and an identification of lattice spacing along that trajectory.

This note isolates the UV requirements needed for a continuum conclusion on a bounded macroscopic observable net. We formulate an explicit *UV package* (trajectory existence, identification, macroscopic scaling limit, and non-collapse of the physical decay rate in physical time units) and show that, once Route 1 provides uniform Euclidean clustering and OS positivity at the macroscopic level, the UV package suffices to produce an OS-positive macroscopic limit functional and a strictly positive Hamiltonian spectral gap on the reconstructed macroscopic theory. All statements are made on bounded macroscopic nets at fixed scale $\ell > 0$ and do not address sharp $t \rightarrow 0$ local fields.

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1 Purpose and scope

Route 1 produces, for four-dimensional lattice Yang–Mills with compact gauge group, a *finite-depth entrance* into a uniform Dobrushin contraction basin at sufficiently large inverse coupling β . Standard consequences are:

- (E1) uniqueness of the infinite-volume lattice state at the blocked scale,
- (E2) exponential Euclidean clustering for bounded local observables, and
- (E3) (by reflection positivity and OS reconstruction) a spectral-gap extraction mechanism on a *bounded macroscopic observable net* at a fixed physical scale $\ell > 0$.

The remaining step, if one wishes to state a *continuum* conclusion, is not part of the entrance mechanism itself: one must specify how $\beta \rightarrow \infty$ is converted into a vanishing lattice spacing $a(\beta) \downarrow 0$ along a tuned ultraviolet (UV) trajectory and how macroscopic observables are identified along that trajectory. This note isolates those UV inputs.

Remark 1.1 (What is and is not addressed). All statements are made on a bounded macroscopic observable net $\mathcal{O}_\ell$ at fixed scale $\ell > 0$. This note does *not* construct sharp local operator-valued distributions obtained by a $t \rightarrow 0$ limit and does *not* solve the UV trajectory/identification problem unconditionally. The goal is to state cleanly what additional UV hypotheses are required to promote the Route 1 entrance/clustering outputs to a macroscopic OS-positive theory with a Hamiltonian spectral gap on $\mathcal{O}_\ell$.

2 Macroscopic nets and induced Euclidean functionals

2.1 A bounded macroscopic net at scale ℓ

Fix a physical scale $\ell > 0$.

Definition 2.1 (Bounded macroscopic net and OS structure). A *bounded macroscopic net at scale ℓ* consists of:

- (N1) For each bounded Borel region $\Lambda \subset \mathbb{R}^4$, a unital $*$ -algebra $\mathcal{O}_\ell(\Lambda)$ of bounded gauge-invariant observables “smeared at scale ℓ ,” with isotony: $\Lambda_1 \subset \Lambda_2 \Rightarrow \mathcal{O}_\ell(\Lambda_1) \subset \mathcal{O}_\ell(\Lambda_2)$. Let $\mathcal{A}_\ell := \bigcup_\Lambda \mathcal{O}_\ell(\Lambda)$.
- (N2) An action τ_x of Euclidean translations $x \in \mathbb{R}^4$ by $*$ -automorphisms of $\mathcal{A}_\ell$ satisfying $\tau_x \mathcal{O}_\ell(\Lambda) = \mathcal{O}_\ell(\Lambda + x)$.
- (N3) A time-reflection $*$ -automorphism ι_θ of $\mathcal{A}_\ell$ induced by $\theta(t, \mathbf{x}) = (-t, \mathbf{x})$ and satisfying $\iota_\theta \mathcal{O}_\ell(\Lambda) = \mathcal{O}_\ell(\theta\Lambda)$. Define the OS reflection as the anti-linear map

$$\Theta(F) := \iota_\theta(F^*).$$

- (N4) The positive-time algebra $\mathcal{A}_{+, \ell} := \bigcup_{\Lambda \subset \mathbb{R}_+^4} \mathcal{O}_\ell(\Lambda)$, where $\mathbb{R}_+^4 := \{(t, \mathbf{x}) \in \mathbb{R}^4 : t > 0\}$.

Remark 2.2 (Continuity conventions). In this note the translation action τ_x is treated algebraically on the bounded net. No intrinsic norm topology on $\mathcal{A}_\ell$ is assumed beyond boundedness of observables. Continuity requirements needed for OS reconstruction are imposed directly on the induced Euclidean functionals, cf. Assumption 3.1(U5).

For this note we do not require a specific realization of $\mathcal{O}_\ell$; it is a target bounded net at macroscopic resolution. In Route 1, $\mathcal{O}_\ell$ is implemented as a bounded gauge-invariant algebra at fixed positive macroscopic scale.

2.2 Lattice approximants and identifications

For a lattice spacing $a > 0$, let $\mathbb{L}_a := a\mathbb{Z}^4 \subset \mathbb{R}^4$. Write $\mu_{\beta,a}$ for an infinite-volume lattice Yang–Mills state at inverse coupling β and spacing a , when it exists. Reflection positivity for the Wilson action (and more generally for standard gauge actions) is classical; see, e.g., [2, 4].

Definition 2.3 (Lattice macroscopic net at scale ℓ). Fix $\ell > 0$ and $a > 0$. For bounded Borel $\Lambda \subset \mathbb{R}^4$, let $\mathcal{O}_{\ell,a}(\Lambda)$ be the unital $*$ -algebra generated by bounded, gauge-invariant cylinder functions of the lattice link field supported on edges of $\mathbb{L}_a$ lying in Λ and of physical diameter $O(\ell)$. Let $\mathcal{A}_{\ell,a} := \bigcup_{\Lambda} \mathcal{O}_{\ell,a}(\Lambda)$. Time reflection and lattice translations act geometrically on $\mathcal{A}_{\ell,a}$ and define an OS reflection $\Theta_a(F) := \iota_{\theta,a}(F^*)$ and positive-time subalgebra $\mathcal{A}_{+, \ell, a}$.

Along a UV trajectory (β_n, a_n) , macroscopic observables in the target net are represented on the lattice by an identification map.

Definition 2.4 (Trajectory identification). Fix $\ell > 0$. A *trajectory identification* consists of maps

$$\iota_n : \mathcal{A}_{\ell} \rightarrow \mathcal{A}_{\ell, a_n}$$

such that:

- (I1) ι_n is a unital $*$ -homomorphism, local in the sense that $\iota_n(\mathcal{O}_{\ell}(\Lambda)) \subset \mathcal{O}_{\ell, a_n}(\Lambda)$ for each bounded Λ .
- (I2) ι_n is compatible with OS reflection on positive time:

$$\iota_n(\mathcal{A}_{+, \ell}) \subset \mathcal{A}_{+, \ell, a_n} \quad \text{and} \quad \iota_n(\Theta F) = \Theta_{a_n}(\iota_n(F)) \quad \forall F \in \mathcal{A}_{+, \ell}.$$

- (I3) For each fixed $F \in \mathcal{A}_{\ell}$, the representatives $\iota_n(F)$ are uniformly bounded in $\|\cdot\|_{\infty}$.

Remark 2.5 (Sufficient condition for translation invariance of $\mathcal{S}_n$). A sufficient condition for translation invariance in Assumption 3.1(U1) is that ι_n intertwines the target translations with the lattice translation automorphisms for shifts in $a_n\mathbb{Z}^4$ (i.e. $\iota_n(\tau_x F) = \tau_{x, a_n}(\iota_n(F))$ for $x \in a_n\mathbb{Z}^4$), together with lattice translation invariance of the underlying state μ_{β_n, a_n} . The UV package in Assumption 3.1 treats translation invariance of $\mathcal{S}_n$ as an explicit macroscopic input.

2.3 Induced Euclidean functionals

Given (β_n, a_n) and ι_n , define the induced macroscopic Euclidean functionals on $\mathcal{A}_{\ell}$ by

$$\mathcal{S}_n(F) := \int \iota_n(F) d\mu_{\beta_n, a_n}, \quad F \in \mathcal{A}_{\ell},$$

whenever μ_{β_n, a_n} exists.

For $F, G \in \mathcal{A}_{\ell}$ and $t \geq 0$, define the *time-separated truncated correlation* under $\mathcal{S}_n$ by

$$\langle F \tau_{(t, \mathbf{0})} G \rangle_{c, n} := \mathcal{S}_n(F \tau_{(t, \mathbf{0})} G) - \mathcal{S}_n(F) \mathcal{S}_n(G).$$

(When $\mathcal{S}_n$ is translation invariant, $\mathcal{S}_n(\tau_{(t, \mathbf{0})} G) = \mathcal{S}_n(G)$.)

3 The UV package

We now state an explicit UV hypothesis package designed to isolate what is external to the Route 1 entrance mechanism.

Assumption 3.1 (UV trajectory and identification package). Fix $\ell > 0$ and a renormalization prescription at scale ℓ . Assume there exist sequences $\beta_n \rightarrow \infty$ and $a_n \downarrow 0$, infinite-volume lattice states μ_{β_n, a_n} , and a trajectory identification ι_n such that:

- (U1) **Macroscopic OS positivity and translation invariance.** For each n , the induced functional $\mathcal{S}_n$ is Euclidean translation invariant on $\mathcal{A}_\ell$ and OS positive:

$$\mathcal{S}_n(\Theta F \cdot F) \geq 0 \quad \forall F \in \mathcal{A}_{+, \ell}.$$

- (U2) **Macroscopic scaling limit on $\mathcal{O}_\ell$.** The functionals $\mathcal{S}_n$ converge pointwise on $\mathcal{A}_\ell$:

$$\mathcal{S}(F) := \lim_{n \rightarrow \infty} \mathcal{S}_n(F) \quad \text{exists for every } F \in \mathcal{A}_\ell.$$

(Equivalently: for every finite collection $F_1, \dots, F_m \in \mathcal{A}_\ell$, the joint law of $(\iota_n(F_1), \dots, \iota_n(F_m))$ under μ_{β_n, a_n} converges; cf. Lemma 4.1.)

- (U3) **UV spacing identification (auxiliary for interpretation).** The map $\beta \mapsto a(\beta)$ defined by the renormalization prescription along the trajectory satisfies $a_n = a(\beta_n)$ and $a_n \downarrow 0$. This item is not used in Theorem 4.4; it is recorded to connect (U4) with lattice-time formulations (Appendix A).

- (U4) **Non-collapse of the physical decay rate (physical time units).** There exist $m_* > 0$ and $N < \infty$ such that for all $n \geq N$ and all $F, G \in \mathcal{A}_\ell$,

$$|\langle F \tau_{(t, \mathbf{0})} G \rangle_{c, n}| \leq C_{F, G} e^{-m_* t} \quad \forall t \geq 0, \tag{1}$$

for constants $C_{F, G} < \infty$ independent of n (but allowed to depend on F, G).

- (U5) **OS reconstruction regularity at the macroscopic limit.** The limiting functional $\mathcal{S}$ satisfies the standard continuity hypothesis needed for OS reconstruction on bounded nets: for all $F, G \in \mathcal{A}_{+, \ell}$, the map $t \mapsto \mathcal{S}(\Theta F \tau_{(t, \mathbf{0})} G)$ is continuous on $[0, \infty)$, where $\tau_{(t, \mathbf{0})}$ denotes time translation in Definition 2.1.

Remark 3.2 (Relation to Route 1 hypotheses). In Route 1, the finite-depth entrance mechanism yields exponential clustering in lattice/block distance at fixed large β . Condition (U4) packages the UV translation of that clustering into *physical* time units along the identification a_n . A standard reformulation in terms of lattice exponential rates is recorded in Appendix A.

Remark 3.3 (Strength of (U4)). Condition (U4) is stated directly in the form needed for OS spectral extraction (Paper 3). It packages (i) a uniform exponential decay rate in physical time and (ii) uniformity in n once F, G are fixed in the macroscopic algebra. It does not require any claim about sharp local fields.

4 Conditional completion theorem

Define the pointwise limit functional $\mathcal{S}$ by Assumption 3.1(U2).

Lemma 4.1 (Joint-law convergence criterion). *Let $F_1, \dots, F_m \in \mathcal{A}_\ell$ be bounded observables and assume that $\mathcal{S}_n(F)$ converges for every F in the unital $*$ -algebra generated by $\{F_1, \dots, F_m\}$. Then the joint law of $(\iota_n(F_1), \dots, \iota_n(F_m))$ under μ_{β_n, a_n} converges as $n \rightarrow \infty$.*

Proof. Each F_i is bounded, so $(F_1, \dots, F_m)$ takes values in a compact subset of $\mathbb{C}^m$. By Stone–Weierstrass, polynomials in the coordinates and their conjugates are uniformly dense in the continuous functions on that compact set. Convergence of $\mathcal{S}_n$ on the generated algebra yields convergence of expectations of all such polynomials, hence convergence of expectations of all continuous bounded test functions and therefore weak convergence of the joint law. $\square$

Proposition 4.2 (OS positivity and translation invariance pass to the macroscopic limit). *Under Assumption 3.1(U1)–(U2), $\mathcal{S}$ is Euclidean translation invariant and Osterwalder–Schrader positive on the bounded macroscopic net $\mathcal{O}_\ell$:*

$$\mathcal{S}(\Theta F \cdot F) \geq 0 \quad \forall F \in \mathcal{A}_{+, \ell}.$$

Proof. Translation invariance is preserved under pointwise limits. For OS positivity, fix $F \in \mathcal{A}_{+, \ell}$. For each n , OS positivity gives $\mathcal{S}_n(\Theta F \cdot F) \geq 0$. Taking $n \rightarrow \infty$ and using pointwise convergence yields $\mathcal{S}(\Theta F \cdot F) \geq 0$. $\square$

Proposition 4.3 (Uniform time clustering passes to the macroscopic limit). *Assume Assumption 3.1(U2) and (U4). Then the limiting functional $\mathcal{S}$ satisfies the same time-separated exponential clustering:*

$$|\langle F \tau_{(t, \mathbf{0})} G \rangle_c| \leq C_{F, G} e^{-m_* t} \quad \forall t \geq 0,$$

for all $F, G \in \mathcal{A}_\ell$, with the same constants $C_{F, G}$.

Proof. For fixed F, G and t , the truncated correlation under $\mathcal{S}_n$ converges to the truncated correlation under $\mathcal{S}$ by (U2) because it is a linear combination of expectations of products of fixed elements of $\mathcal{A}_\ell$. The bound (1) is uniform in n , hence passes to the limit. $\square$

Theorem 4.4 (Conditional completion on a bounded macroscopic net). *Assume Assumption 3.1(U1), (U2), (U4), and (U5). Then the macroscopic limit functional $\mathcal{S}$ is OS positive and translation invariant on $\mathcal{O}_\ell$, admits OS reconstruction on the bounded net, and the reconstructed Hamiltonian H has a strictly positive spectral gap:*

$$\text{Spec}(H) \subset \{0\} \cup [m_*, \infty).$$

Proof. By Propositions 4.2 and 4.3, $\mathcal{S}$ satisfies OS positivity and exponential time clustering on the bounded macroscopic net. By Assumption 3.1(U5) and the standard OS reconstruction theorem (see, e.g., [1, 3, 4, 5]), $\mathcal{S}$ yields a Hilbert space $\mathcal{H}$, a distinguished vector Ω , and a strongly continuous contraction semigroup e^{-tH} implementing time translations.

The exponential time clustering bound implies an exponential decay estimate on semigroup matrix elements for centered vectors generated by macroscopic observables. By the semigroup spectral extraction criterion (Paper 3, Theorem 5.1), spectrum in $(0, m_*)$ is excluded and hence the spectral gap bound holds. $\square$

Remark 4.5 (Terminology and vacuum sector). The conclusion is a *Hamiltonian spectral gap* on the reconstructed macroscopic theory at fixed scale $\ell > 0$. No claim is made here about sharper ultraviolet localization (e.g. existence of pointlike fields) or about the full relativistic energy–momentum spectrum. Depending on the precise OS reconstruction package invoked, one may phrase the conclusion as a gap above the ground-state sector; in the common OS setup with a cyclic vacuum vector, exponential clustering excludes spectrum in $(0, m_*)$ and yields $\text{Spec}(H) \subset \{0\} \cup [m_*, \infty)$.

5 Discussion: what is external and what is internal

The Route 1 entrance mechanism is an internal, finite- β statement: after a computable blocking depth, the exact blocked specification is placed in a uniform contraction basin. The UV package isolates what is external to that mechanism:

- existence of a tuned trajectory $\beta_n \rightarrow \infty$ with an identification $a_n \downarrow 0$ selected by a renormalization prescription;
- existence of the macroscopic scaling limit on the bounded net $\mathcal{O}_\ell$;
- a non-collapse condition translating lattice/block exponential rates into a strictly positive *physical* decay rate.

The purpose of this separation is not rhetorical. It makes clear which parts of the macroscopic mass-gap conclusion follow from the entrance mechanism and OS spectral theory, and which parts amount to the general UV completion problem for four-dimensional Yang–Mills.

Remark 5.1 (Why the UV package is formulated on bounded nets). The macroscopic algebra $\mathcal{O}_\ell$ is chosen to avoid ultraviolet localization demands beyond fixed scale $\ell > 0$. The OS positivity and spectral-gap extraction are therefore formulated entirely at the level of bounded observables at that scale. Sharp $t \rightarrow 0$ local fields correspond to a strictly stronger ultraviolet regularity problem, not treated here.

A A convenient reformulation of (U4)

Remark A.1 (Role of (U3)). This appendix is interpretive and is not used in the proof of Theorem 4.4. The spacing-identification item (U3) enters only to relate the physical-time formulation (U4) to a lattice-time decay statement in the standard way.

Route 1 typically yields exponential clustering in *lattice* units of the form

$$|\langle A \tau_{ke_0} B \rangle_{c,n}| \leq C_{A,B} e^{-m_{\text{lat}}(\beta_n)k}, \quad k \in \mathbb{N},$$

for bounded observables A, B supported in a fixed number of lattice spacings and with e_0 the unit vector in the Euclidean time direction. If physical time t corresponds to $k = t/a_n$ lattice steps along the UV identification, then

$$e^{-m_{\text{lat}}(\beta_n)k} = e^{-(m_{\text{lat}}(\beta_n)/a_n)t}.$$

Thus (U4) is equivalent to the existence of $m_* > 0$ such that

$$\liminf_{n \rightarrow \infty} \frac{m_{\text{lat}}(\beta_n)}{a_n} \geq m_*,$$

provided the identification intertwines physical and lattice time translations in the following explicit sense: for each n , each $k \in \mathbb{N}$ and $t := ka_n$, and each $F \in \mathcal{A}_\ell$,

$$\iota_n(\tau_{(t, \mathbf{0})} F) = \tau_{ke_0, a_n}(\iota_n(F)),$$

where τ_{ke_0, a_n} denotes lattice translation by k steps in the Euclidean time direction on $\mathcal{A}_{\ell, a_n}$. This is the form recorded in Route 1 as “non-collapse of the physical mass scale.”

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